

Math 2650 – Quiz 02 Concept Practice

Choose the section matching the skill you want to practice. Answers are on the back.

A. Planes and Perpendicular Geometry

Let $P = (1, 1, 0)$ and $Q = (5, -1, 4)$, and let X be the set of all points equidistant from P and Q .

- (a) Find an equation of the plane X .
- (b) Let M be the midpoint of $\overline{PQ}$, and let ℓ be the line through M with direction vector $\langle 1, 2, 0 \rangle$. Find an equation of the plane Y that contains ℓ and is perpendicular to X . Also give a unit normal vector to Y .
- (c) Find the point on X closest to the origin.

B. Triangle Geometry and Cross Product

A triangle has vertices

$$U = (0, 0, 0), \quad V = (4, 0, 0), \quad W = (1, a, 2a).$$

- (a) Find the area of the triangle in terms of a .
- (b) Find all values of a for which the area is $6\sqrt{5}$.

C. Projections and Parallel Lines

Consider the parallel lines

$$L_1: \quad x = 2 + t, \quad y = 2 + 2t, \quad z = 3 + t,$$

and

$$L_2: \quad x = 5 + s, \quad y = 7 + 2s, \quad z = 2 + s.$$

Let $A = (2, 2, 3)$ be the point on L_1 corresponding to $t = 0$, and let $B = (5, 7, 2)$ be the point on L_2 corresponding to $s = 0$.

- (a) Find the projection of $\overrightarrow{AB}$ onto the direction of the lines.
- (b) Find the distance between L_1 and L_2 .

D. Angles and Unit Vectors and Dot Product

Let θ be the angle between vectors $\mathbf{u}$ and $\mathbf{v}$. Suppose

$$|\mathbf{u}| = 3, \quad |\mathbf{v}| = 5, \quad (\mathbf{u} + 2\mathbf{v}) \cdot (3\mathbf{u} - \mathbf{v}) = 27.$$

- (a) Find $\cos \theta$.
- (b) Let $\mathbf{q} = \langle 2, 1, 2 \rangle$. Find all unit vectors $\mathbf{w} = \langle a, 0, b \rangle$ that make the same angle θ with $\mathbf{q}$.

Answer Key

Final answers only

A. Planes and Perpendicular Geometry

(a)

$$2x - y + 2z = 10$$

(b)

$$4x - 2y - 5z = 2$$

and one unit normal vector is

$$\frac{1}{3\sqrt{5}} \langle 4, -2, -5 \rangle.$$

(c)

$$\left(\frac{20}{9}, -\frac{10}{9}, \frac{20}{9} \right)$$

B. Triangle Geometry

(a)

$$2\sqrt{5}|a|$$

(b)

$$a = \pm 3$$

C. Projections and Parallel Lines

(a)

$$\text{proj}_{\langle 1, 2, 1 \rangle} \overrightarrow{AB} = \langle 2, 4, 2 \rangle$$

(b)

$$d(L_1, L_2) = \sqrt{11}$$

D. Angles and Unit Vectors

(a)

$$\cos \theta = \frac{2}{3}$$

(b)

$$\mathbf{w} = \langle 1, 0, 0 \rangle \quad \text{or} \quad \mathbf{w} = \langle 0, 0, 1 \rangle$$